UNIVERSITY OF THE
WITWATERSRAND,
JOHANNESBURG

HREC APPLICATION FORM (MEDICAL) - 2022

UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG

APPLICATION [1] TO THE HUMAN RESEARCH ETHICS COMMITTEE: (MEDICAL) - FOR CLEARANCE OF RESEARCH – for Pharmaceutical / Grant / Donor Sponsored Clinical Trials involving Drugs / Devices

THIS APPLICATION MUST BE TYPED
Please complete ALL Investigators' details involved in the Study as requested below (including Item 6.3):

SECTION 1

1 PRINCIPAL INVESTIGATOR PER SITE
NAME: Prof Matthew Francis Chersich

- **PROFESSIONAL STATUS:** Research Professor
- **UNIVERSITY DEPARTMENT**: Wits RHI, Faculty of Health Sciences
- **DETAILS OF SITE WHERE STUDY WILL BE CONDUCTED / (SYNDICATE NAME):** Wits RHI
- **HOSPITAL / INSTITUTION WHERE EMPLOYED**: Wits RHI
- **FULL-TIME OR PART-TIME:** Full-time **HPCSA NO:** MP0508365
- **TELEPHONE NO:** +27 11 358 5300
- CELL:+27 72 752 1123 **EMAIL:** mchersich@wrhi.ac.za

OR

1.1 NON-WITS SITE / INSTITUTION *(if no association of any type with the University) e.g. Private Research Site; Laboratories; NA*

- DETAILS OF SITE WHERE STUDY WILL BE CONDUCTED
  Full Address
- TELEPHONE NO: FAX NO:
- CELL: EMAIL:

GOOD CLINICAL PRACTICE (GCP)
GCP TRAINING: Please list Principal and all Sub-Investigators GCP Training
Please indicate DATE AND NAME of GCP Course attended (dd/mmm/year)
(Investigators meetings do not qualify as GCP training)
(Please attach study specific CV's AND Declarations in the required format.)

Full Name: Matthew Francis Chersich
GCP Course Name: SA GCP 2020
Date of GCP course: day/month/year: 18 May 2022

REQUIREMENT FOR ETHICS TRAINING:
Following the audit and recommendation made by The National Health Research Ethics Council (NHREC) in July 2018, ethics training is now a compulsory requirement for consideration of this application (in addition to Good Clinical Practice Training).

NB: If ethics training has not been done yet, you may still submit the application and once ethics training has been completed, submit a copy of the ethics training certificate.

Please refer to the Wits SOP on Ethics Training, as well as the Ethics in Health Research: Principles, Processes and Structures, Department of Health 2015 Guidelines, Second Edition, Appendix 2, page 80-81, for free online ethics training opportunities:

Follow the link: https://www.witshealth.co.za/Services/Research-Ethics/Application-Forms - click on item 26

See example of an ethics training course: TRREE (ONLY MODULES 1 AND 3.1 REQUIRED): https://elearning.trree.org/course/index.php?categoryid=1

Full Name: Matthew Francis Chersich
Ethics Course Name: TRREE Introduction to Research Ethics and Informed Consent
Date of Ethics Course: day/month/year: 27 May 2022

SECTION 2

2 PLEASE SUPPLY CONTACT PERSON, ADDRESS AND CONTACT DETAILS TO WHOM ALL CORRESPONDENCE SHOULD BE ADDRESSED:

- **NAME:** Darshnika Pemi Lakhoo
- **COMPANY:** Wits RHI
- **ADDRESS:** Hugh Solomon Building, 22 Esselen Street, Hillbrow, Johannesburg 2001
- **TELEPHONE NO:** +2711 358 5300
- **CELL:** +27 82 564 3595 **EMAIL:** dlakhoo@wrhi.ac.za

SECTION 3

3. PROTOCOL / PROJECT NUMBER / STUDY NUMBER: HEAT001

VERSION / AMENDMENT NUMBER: Version 1.0

DATE: 05 May 2022

3a. TITLE OF RESEARCH PROJECT: Individual Participant Data meta-analysis to quantify the impact of high ambient temperatures on maternal and child health in Africa

WHERE WILL THE RESEARCH BE CARRIED OUT?
(Please furnish name of the Research Unit / Site and/or Hospital/Institution and particular department – if applicable)

Wits Reproductive Health and HIV Institute
Hugh Solomon Building
22 Esselen Street
Hillbrow
Johanneburg
2001

3b. HREC PRINCIPLES AND POLICY ON BIOBANKS:

If this project involves the use of a Biobank, then this is dealt with by the Biobanks Ethics Committee (BEC), a Subcommittee of the HREC (Medical):

BEC requires application forms to be submitted for:

- Approval of a Biobank for sample storage (BEC 1 Form) or for
- Storage/Retrieval/Transfer of HBM in a BEC approved Biobank (BEC Form 2) – (Please follow the link below for the latest list of approved biobanks)

Will this study involve the use of a Biobank? **Yes** ☐ **No** ☒

If yes, please submit a parallel submission to BEC. Please contact Rhulani Mkansi (Rhulani.Mkansi@wits.ac.za / 011 717 1234/1252) for more information on the directives, guidelines and procedures of BEC.

Please follow the link below for submission requirements to the BEC:
https://www.wits.ac.za/research/researcher-support/research-ethics/ethics-committees/

All the following sections must be completed [3]. **Please tick all relevant boxes.**

3.1 PURPOSE OF THE RESEARCH:

Postgraduate: degree/diploma (state which) ☐

Undergraduate: degree/diploma (state which) ☐

Not for degree purposes - (Commercial Trials) ☒

3.2 OBJECTIVES OF THE RESEARCH (please list): Please do not refer to an attached list – expand the space below to include the information required:

Primary objective

1. To use innovative data science approaches to quantify the current and future impacts of heat exposure on maternal and child health in sub-Saharan Africa.

Secondary objectives

Secondary objectives are:

1. To acquire, collate and integrate prospectively collected data from cohort studies and randomized trials on maternal and child health in sub-Saharan Africa
2. To develop a collaboration between study groups and the IPD investigators
3. To link health outcome data spatially and temporally with weather and other environmental data, as well as socio-economic and other data
4. To utilize classic statistical and novel machine learning approaches to understand and quantify the impact of heat exposure on pregnant women and children
5. To document in detail the variations in the relationship between heat exposure, and maternal and children health outcomes across different settings, climate zones and population sub-groups

3.3 SUMMARY OF THE RESEARCH (give a brief outline of the research plan): Please do not refer to an attachment – expand the space below to include the information required:

Rationale: Globally, recognition is growing of the harmful impacts of extreme heat on health, across a range of population groups, including pregnant women and children. There remain, however, major gaps in evidence on the extent to which high temperatures increase the risks for adverse health outcomes and which outcomes are most affected. Gaps in evidence are especially large in Africa. An individual participant data (IPD) meta-analysis entails systematically locating, appraising, transforming, and analysing individual data from multiple studies which have a common outcome of interest. Therefore, unlike classical systematic reviews, the IPD uses participant-level data, rather than simply using aggregate-level data extracted from a publication. This approach can overcome many of the biases of classic systematic reviews of published data, and the challenges in understanding heterogeneity in findings, and methodological diversity across published studies. Additionally, analysing pooled participant-level data from multiple settings and time periods can mitigate many of the limitations of analysis that include individual databases from a single location and time, most especially in limitations of study power and generalisability. A detailed understanding and quantification of the linkages between heat and health is essential for developing solutions to this critical policy and research area.

The study forms parts of DS-I Africa which aims to make optimum use of existing data resources across Africa to address the most pressing health concerns on the continent. The study constitutes one of two research projects within the HEat and HEalth African Transdisciplinary Center (HE²AT Center) project funded through the DS-I Africa Program.

Study Design: Systematic review and meta-analyses of individual participant data.

Methods: We will systematically identify eligible studies through a mapping review, searching through data repositories, as well as suggestions from experts and investigators in this field. Inclusion is based on study- and individual-level criteria. To be eligible, studies need to have collected longitudinal data, have enrolled at least 1000 pregnant women from sub-Saharan Africa in the period January 2000 to June 2022, have collected data on key outcomes of the review, and have relevant ethics approvals in place. At an individual level, participants need to have been recruited during pregnancy or intrapartum and have data available on date of birth and location. We will obtain data from open sources, data repositories or through a collaboration with investigators or data holders from eligible studies. The individual datasets will be harmonised through the recoding of raw individual participant data into a common set of variables. A one and/or two-stage analysis method will be adopted whereby, in the first-stage, each study is analysed individually. Then, in the second stage, the data from the individual studies are aggregated to provide a pooled estimate of effect. If heterogeneity between studies is high, then aggregation across studies may not be done, or only be done in particular groups of studies that share common characteristics and have low heterogeneity. Various traditional models such as time-series analysis, time-to event analysis and generalised additive models as well as novel machine learning approaches will be used to quantify associations between exposure to high ambient temperatures, and adverse maternal and child outcomes.

3.4 What are the Investigator's anticipated ethical concerns with the project?

The project raises several ethical and legal considerations. This includes 1) the use of secondary data from pregnant women and children for research purposes that may differ from the original purpose of the study, 2) sharing data across countries in Africa, and 3) the presence of potential identifying information.

3.5 What capacity building is planned for South Africa in this study?

This project falls within a wider consortium (HE²AT Center) that has a designated training and engagement core (TEC). Through the TEC, the HE²AT Center will serve as a Center of Excellence and resource in Africa. The HE²AT Center will engage with the communities in which the research is conducted, to increase acceptability of the study and its objectives. These activities use existing engagement mechanisms, engaging with local Community Advisory Boards and media and distributing newsletters. Because the solutions developed by the Center address two major concerns for

policymakers (how to warn people about a heatwave and then to track its impacts), we anticipate that local, national and international policymakers will be willing to engage. The TEC aims to:

- Advance knowledge and skills of early career researchers and scientists of the HE²AT Centre and other Research Hubs on transdisciplinary heat-health research using data science approaches
- Build knowledge, skills, and competencies in solutions for addressing heat-health impacts, particularly early warning and response systems
- Develop and implement a systematic strategy for forming, maintaining, and strengthening engagement with governments, industry, non-governmental and related organizations, and communities
- Disseminate results through scientific publications and presentations at scientific meetings, community engagement, policy engagement, and reports to NIH.

Many of these activities will be conducted in South Africa with extension into the wider study group in other parts of Africa.

SECTION 4

4. REQUIREMENTS

4.1 If this project involves studies with drugs at a teaching hospital associated with this University, approval must first be obtained from the Protocol/Expert Reviewer.
Has application been made? **Yes** ☐ No ☐
(If not, this application cannot be considered)

NA

4.2 In terms of radiation aspects of radiological investigations or use of isotopes related to the trial:

4.2.1 Please confirm if the radiological investigations or use of isotopes required in the study are in line with accepted standards of care for the patient's condition?
NA

a. If Yes, please provide a written letter from a non-study related clinician that is able to confirm this
NA

4.2.2 If No, please advise if the study has already been approved in Europe; and/or US and/or Japan:
NA

a. For studies already approved in Europe please list the European countries in which the study has been approved and when:

b. For studies not yet approved in Europe but approved in the US and/or Japan please provide the details of the expert review body (or bodies) and a list of their members that considered and approved the radiation risk benefit of the imaging required in the protocol:

4.2.3 If the study has not yet been approved in any of the above jurisdictions written approval must be obtained from the Director, Radiation and Health Physics Unit (james.larkin@wits.ac.za / 011 717 6931). For patients these are radiation dosages over and above those for standard diagnostics / therapy

NA

Is this attached? If not, the application cannot be considered. Yes ☐ No ☐
NA

4.3 Participant Information Sheet[4] is attached. (For written and verbal consent)
NA Yes ☐ No ☐

Informed Consent Form[5] is attached. (For written consent). Yes ☐ No ☐
Please refer to Wits Informed Consent Form Template – www.witshealth.co.za – select Ethics for template:
NA

Consent will be verbal NA Yes ☐ No ☐

Informed consent is not necessary. State why not. Yes ☒ No ☐
No new consent for our study will be conducted. We will be using secondary data. The informed consent process for each dataset that agrees to contribute data to the IPD will be assessed for ethical considerations. Firstly, we will review informed consent procedures that were undertaken by the participant for the primary study. If a participant has signed a "broad consent" for use of their data in future research projects, even though the precise nature of the future research is unclear, this will allow data sharing without further ethical approvals.
Participants that have signed "narrow consent", where sharing of data is not permitted, and/or "tiered consent", where each individual participant chooses to permit data sharing or not, will be carefully considered. The possibility for reconsenting will be deliberated if no consent for data sharing (narrow and tiered with no consent) has been provided by the participant. If reconsenting is not feasible, impossible, or would involve a disproportionate effort, an informed consent waiver will be requested from the UCT Faculty of Science Research Ethics Committee.

4.4 If a Questionnaire or Interview is to be used in the research, it must be attached. Is it attached? *If not, the application cannot be considered.* Yes ☐ No ☐
NA

SECTION 5

5. PARTICIPANTS FOR STUDY

5.1 If patients are being studied, state where and how the participants are selected: *(i.e. recruitment strategies)*
Participants are selected from individual participant data from trials and cohorts that have already been conducted (Secondary data analysis)

The process below describes the eligibility criteria for the studies and the participants through the induvial participant data meta-analysis methodology.

Eligibility is determined at the study level and at an individual level, where a set of criteria determine the eligibility of individual participants within the eligible studies.

Eligibility criteria at the study level and rationale

The IPD includes studies that collected longitudinal data from medium-to-large cohorts and clinical trials. Longitudinal data allows for the assessment of temporal trends and the use of statistical approaches like time-to-event analyses which are valuable in assessing heat-health associations. Including data from longitudinal studies where women are followed over pregnancy in one place helps avoid biases in birth registries where women may have spent much of her pregnancy in a place some distance from the site where she had childbirth.

The study only includes cohorts/trials that enrolled more than 1000 participants given that the large amount of time and resources required for data acquisition, preparation, and analysis of each individual study, makes it difficult to justify the inclusion of smaller studies.

The following criteria are required for a **study to be eligible for inclusion**:

1. Cohort or trial that enrolled at least 1000 pregnant women in a study site(s) in sub-Saharan Africa, published between January 2000 and 30 June 2022
2. Located through published literature in the systematic mapping, clinical trial registry or other means
3. The study can be single- or multi-site, and across one or more countries in Africa, or other parts of the world
4. Study design is a randomized or non-randomised clinical trial, or an observational or interventional cohort with prospectively collected data
5. At least one primary outcome or two of the 'important' maternal and neonatal/child have been collected as part of the study
6. The study had obtained the relevant local ethics approvals and any concerns around ethical procedures have been clarified by the data provide

The following are **exclusion criteria** at the study level:

1. Systematic and narrative reviews, reports of birth registries, cross-sectional studies, health and demographic surveillance data, or case control studies

Eligibility criteria at the participant level

Even though a study may fulfil the eligibility criteria, some individuals within the study may be ineligible. We apply a set of eligibility criteria for individuals within a study.

The following criteria need to be met for a ***participant (woman and her child)*** to be eligible for inclusion:

1. Be a participant from an eligible study
2. The woman was enrolled into the study either during pregnancy or intrapartum
3. Individual participant information is at least available on date of birth of the newborn, or date of end of pregnancy in cases of maternal deaths or abortion
4. Individual participant information is at least available on location of birth, or location of study follow-up (city-level or more granular)

5.2. Where the participants are not patients,
They will be invited to volunteer ☐ **they will be selected** ☐
***State how the participants are selected, or who is invited to volunteer*:** NA

Are the participants subordinate to the person doing the recruiting? Yes ☐ No ☒
***If yes, justify the selection of subordinate participants*:**

5.3 Will control participants be used? Yes☐ No☒
If yes, explain how they will be recruited – for clinical trials this refers to Control Arm

5.4 Participants records: state what records will be used and how they will be selected:
Individual participant data will be used from cohorts and clinical trials that have already been conducted.

5.5 Age range of patients/participants/controls:

If under 18 years, from who will consent be obtained?
If consent is required for re-use of data for this study, consent will be obtained as it was detailed in the original, approved study.

If under 18 years, is a Patient Assent Document included? NA
Yes☐ No☐

5.6 Sex: Male ☒ **Female** ☒

5.7 Number of patients _____ ***Applicable to clinical trials***
Non-patient participants _____ ***Not applicable to commercial clinical drug trials***
Controls _____ ***Not applicable to commercial clinical drug trials***

It is not certain what number of participants we will obtain for this study.

5.8 Benefit to patient or participants: will the research benefit the patient(s) or participant(s) in any direct way. If yes, explain in what way. Yes ☐ No ☒

Is the patient being remunerated for participating in the study? Yes☐ No☒
If yes, please state what the remuneration is for and how much will be paid. _____

5.9 Disadvantages to patients/participants/controls. Will participation or non-participation disadvantage them in any way? If yes, explain in what way:
Yes ☐ No ☒

5.10 Describe what post-trial access to test medicament (medication) there will be to participants who respond well to the test medicament shown by objective measurements. NA

5.11 Will participants in the control arm also receive post-trial access to test medication?
NA

SECTION 6

6. PROCEDURES

6.1 Mark research procedure(s) that will be used:

☐ **Record review (patient file)**

☐ **Interview form (must be attached)**

☐ **Questionnaire (must be attached)**

☐ **Examination (state below nature and frequency of examination)**

☐ **Drug or other substance administration (state below name(s) of drug(s)/substance(s) and dose(s) and frequency of administration)**

☐ **X-rays**

☐ **Isotope administration (state below name(s) of isotope(s) and frequency)**

☐ **Blood sampling;** ☐**venous;** ☐**arterial (state below amount to be taken and the frequency of blood sampling)**

☐ **Biopsy**

Please note: Should a study require a portion or slides of histological diagnostic specimens to be sent away for testing, the anatomical pathologist who has custody of the specimen block must agree to this in writing.

☒ **Other procedures (explain)**

Use this space to elaborate procedures marked above:

This study will use individual participant data from clinical trial and cohort study records. It is a secondary analysis of existing data from trials and cohorts. No new data will be collected. No new samples will be collected. No new procedures will be performed.

6.2 Is / are procedure/(s) routine for: Diagnosis / Management? ☐
Specific to research? ☒

6.3 Who will carry out the procedure(s)?

- **(State name/(s) and position/(s) held per research unit.**

Matthew Chersich – Research Professor
Craig Parker – Data scientist
Ijeoma Solarin – Researcher
Gloria Maimela – Director of Climate and Health at Wits RHI
Zororo Mavindidze – Portfolio manager
Darshnika Lakhoo – Researcher
Lee Fairlie – Director of Maternal and Child Health Wits RHI

- **All Principal and Sub-investigators must be listed here,**
- **Please attach the study specific CV's AND Declarations in the required format.**

GCP TRAINING: Please list all Investigators GCP Training
Please indicate date and name of GCP Course attended
(Investigators meetings do not qualify as GCP training)

Full Name: Matthew Francis Chersich
GCP Course Name: SA GCP 2020
Date of GCP course: day/month/year: 18 May 2022

Full Name: Craig Parker
GCP Course Name: SA GCP 2020
Date of GCP course: day/month/year: 16 May 2022

Full Name: Gloria Maimela
GCP Course Name: SA GCP Basic Online Course
Date of GCP course: day/month/year: 08 September 2020

Full Name: Darshnika Pemi Lakhoo
GCP Course Name: SA GCP Basic Online Course
Date of GCP course: day/month/year: 13 October 2020

Full Name: Ijeoma Solarin
GCP Course Name: SA GCP 2020
Date of GCP course: day/month/year: 17 May 2022

Full Name: Lee Fairlie
GCP Course Name: SA GCP 2020
Date of GCP course: day/month/year: 20 April 2021

(It is a requirement that a GCP course be attended every THREE years)

ETHICS TRAINING: Please list all Investigators Ethics Training
Please indicate date and name of Ethics Training Course completed

Full Name: Matthew Francis Chersich
Ethics Course Name: TRREE Introduction to Research Ethics and Informed Consent
Date of Ethics Course: day/month/year: 27 May 2022

Full Name: Craig Parker
Ethics Course Name: TRREE Introduction to Research Ethics and Informed Consent
Date of Ethics Course: day/month/year: 17 May 2022

Full Name: Gloria Maimela
Ethics Course Name: TRREE Introduction to Research Ethics
Date of Ethics Course: day/month/year: 09 September 2020

Full Name: Gloria Maimela
Ethics Course Name: TRREE Informed Consent
Date of Ethics Course: day/month/year: 11 September 2022

Full Name: Darshnika Pemi Lakhoo
Ethics Course Name: TRREE Introduction to Research Ethics and Informed Consent
Date of Ethics Course: day/month/year: 07 October 2020

Full Name: Darshnika Pemi Lakhoo
Ethics Course Name: ACRP Ethics and Human Subject Training: A Comprehensive Introduction
Date of Ethics Course: day/month/year: 06 September 2021

Full Name: Ijeoma Solarin
Ethics Course Name: TRREE Introduction to Research Ethics
Date of Ethics Course: day/month/year: 15 April 2022

Full Name: Ijeoma Solarin
Ethics Course Name: TRREE Informed Consent
Date of Ethics Course: day/month/year: 16 April 2022

Full Name: Lee Fairlie
Ethics Course Name: CITI Program Human Subject Protection
Date of Ethics Course: day/month/year: 19 April 2021

Full Name: Zororo Mavididze
Ethics Course Name: TRREE Introduction to Research Ethics and Informed Consent
Date of Ethics Course: day/month/year: 24 May 2022

6.4 Who will be collecting data, blood samples etc.
We will not be collecting any new data. Secondary data will be collected by all study staff.

6.5 Are data collectors part of the study team or service providers?
Yes, all study staff will assist with reaching out to investigators to request data.

6.6 Are data collectors aware of and have they agreed to collect data / specimens?
Yes

6.7 When will the research commence in South Africa and/or internationally, and over what approximate time period will the research be conducted?

START DATE: September 2021 **END DATE:** 30 June 2026

RECRUITMENT START: 30 June 2022 **END DATE:** 30 June 2026

Please include: Anticipated number of participants to be enrolled at each site.
NA

6.8 **Research on studies being conducted outside the Gauteng Academic Hospitals, please list the number of other studies currently being conducted by the Principal Investigator, the number of patients per study and state where they are being performed.**

The principal investigator is involved in two other observational studies - they are not responsible for any patients.

6.9 **For applications outside the Gauteng Academic Hospitals: Is the investigator involved in clinical trials – in a Part-Time / Full-Time capacity (delete whichever is not applicable)**

The principal investigator is not involved in clinical activities.

SECTION 7

7. **RISKS OF THE PROCEDURE(S) participants/controls will suffer:**

☐ **No risk** ☐ **Discomfort**
A. Routine diagnosis and standard of care
B. Over and above those for routine diagnosis and treatment

☐ **Pain** ☒ **Possible complications**

☐ **Side effects from agents used**

If you have checked any of the above except "No risk" provide details here: There is a minimal risk of breech of confidentiality. Risk to a participant in this study are minimal as there are no new procedures involved and many steps will be taken to protect their personal information. Data from participants will be safeguarded and only made available to study investigators through strict access control platforms. Personal identifying information such as date of birth or address will further, only be made available to a smaller subset of investigators who will be responsible for anonymising the data through various methods. Moreover, no personal details of any participant will be made publicly available through this project or published in a journal article.

SECTION 8

8. **GENERAL**

8.1. **Has permission of relevant authority/ies been obtained?** **Yes**☐ **No**☐ **N/A**☒
State name of authority/ies:

8.2 **Has this study been submitted to other Ethics Committees?** **Yes**☐ **No**☒ **N/A**☐
If yes, what is the status of the application?
The study will be submitted to UCT Science Ethics Committee – the submission is pending

8.3 **Confidentiality:**

Please explain how confidentiality will be maintained so that participants / controls are not identifiable to persons not involved in the research by answering the questions below:

i. Data de-identification?
ii. Data access / protection (data security)?
iii. Period of data storage?
iv. Period of sample storage?
v. Destruction of data/samples
vi. Sample transfer (Material Transfer Agreements)?
vii. Data transfer (Data Transfer Agreements)?

Data may not be fully unidentifiable as we are collecting date of birth and location data, where available. However, we will take steps to minimise the risk to privacy of participants. We will not be collecting names of participants and no identifiable data will be published. In addition, the data will be safeguarded in a password protected server with limited access. In addition, we will use principles of minimisation, to ensure that only essential data that is required for the study objectives are stored. Lastly, wherever possible, we will anonymise data through three methods:

1. Street addresses will be aggregated into geographical regions such that many records map to the same region and it is no longer possible to derive individual residential locations. Aggregation will also consider the number of records that map to the same geographical area.
2. Latitude/longitude coordinates will be "jittered" which involves adding random values to each coordinate such that the exact location is lost, while retaining sufficient geographical information to support the analysis. Jittering will be used to shift latitude/longitude locations in the order of 10s of kilometres, adequate to prevent locating individual residential locations.
3. Where Date of Delivery/Date of Birth is considered personally identifiable information, this will be used to associate the relevant climate variables and then will be removed from the dataset.

8.4 Results: to whom will result be made available?

DSI-Africa consortium, policymakers, the scientific community, the general public, local, provincial, and national authorities and other stakeholders.

8.5 Finances: There will be financial costs to:

Patient/Participants	**Yes☐**	**No☒**
Hospital/institution	**Yes☐**	**No☒**
Other	**Yes☐**	**No☒**

Explain any box marked "Yes":

8.6 How will the research be funded?

Please complete details of Funder / Donor / Sponsor / Pharmaceutical Company - (Full name and address)

The research is funded by the Fogarty International Center and National Institute of Environmental Health Sciences (NIEHS) and OD/Office of Strategic Coordination (OSC) of the National Institutes of Health under Award Number U54 TW 012083.

8.7 Any other information, which may be of value to the Committee, should be provided here:

NA

8.8 Select DAIDS Risk/Benefit Category (If NIH Sponsored):

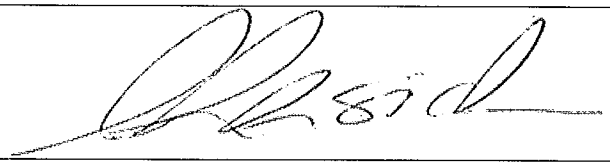
Subpart D – Additional Protections for Children Involved as Participants in Research:

X	Category	Description
☒	45 CFR 46.404	Research not involving greater than minimal risk.
☐	45 CFR 46.405	Research involving greater than minimal risk but presenting the prospect of direct benefit to the individual subjects.
☐	45 CFR 46.406	Research involving greater than minimal risk and presenting no prospect of direct benefit to individual subjects, but likely to yield generalizable knowledge about the subject's disorder or condition.
☐	45 CFR 46.407	Research not otherwise approvable which presents an opportunity to understand, prevent, or alleviate a serious problem affecting the health or welfare of children.
☐	45 CFR 46.408	Requirements for permission by parents or guardians and for assent by children.
☐	45 CFR 46.409	Wards

The Ethics Assessors will select a DAIDS Risk/Benefit Category for NIH Sponsored studies that include minors.

APPLICANT'S INFORMATION AND SIGNATURE:

Note: THE PRINCIPAL INVESTIGATOR IS THE APPLICANT

Name: *Please Print Name*	**Matthew Francis Chersich**
Department:	**Climate and Health Directorate, Wits RHI**
Telephone No:	**+27 752 1123**
Date	**10 June 2022**
Signature:	

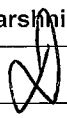
WHO WILL SUPERVISE THE PROJECT? May not be applicable to commercial clinical trials / drug trials / device trials– please insert N/A – not applicable)

Name: *Please Print Name*	**NA**
Department:	
Telephone No:	
Date	.
Signature:	

HEAD OF DEPARTMENT / UNIT OF INSTITUTION IN WHICH STUDY WILL BE CONDUCTED

Name: *Please Print Name*	Professor Helen Rees
Signature:	
Date:	14 June 2022
Head of Dept / Unit of Institution where the Study will be conducted:	Wits RHI
Tel No:	011 358 5344
Fax No:	N/A
Email:	hrees@wrhi.ac.za

NAME, CONTACT DETAILS AND SIGNATURE OF COMPANY REPRESENTATIVE / RESEARCH COORDINATOR:

Name: *Please Print Name*	**Darshnika Pemi Lakhoo**
Signature:	
Date:	**10 June 2022**
Company:	**Climate and Health Directorate, Wits RHI**
Tel No:	**+27 82 564 3595**
Fax No:	
Email:	**dlakhoo@wrhi.ac.za**

PLEASE NOTE:

1 **If not employed by the University or one of the University's teaching hospitals, please indicate clearly, where correspondence should be sent.**

2 **This requirement holds even if, to assist the Committee, a protocol detailing the background to the research, the design of the investigation and all procedures, is submitted with the application.**

3 **NB. If any doubt exists please contact WHC Ethics Secretariat to the Human Research Ethics Committee: (Medical)**
At 011 274 9278 / 9279 / 9280 / 9255 / 3353 – Email: EthicsRegulatory@witshealth.co.za

4 **Whether written or verbal consent is to be obtained, the HREC requires a Participant Information Sheet written in language understandable to the participant (or guardian) detailing what the participant will be told. This should include the following:**

(1) **Participation is voluntary, and refusal to participate will involve no penalty or loss of benefits to which the participant is otherwise entitled:**
(2) **The participant may discontinue participation at any time without penalty or loss of benefits:**
(3) **A brief description of the research, its duration, procedures and what the participant may expect and/or be expected to do:**
(4) **Any foreseeable risks, discomforts, side effects or benefits, including those for placebo:**
(5) **Disclosure of alternatives available to the participant. If risks are involved:**
(6) **A professional contact name and 24 hour telephone number:**
(7) **Explanation whether medical treatment will be provided in the case of a complication developing:**
(8) **Compensation for trial related injuries will be in accordance with the ABPI guidelines:**
(9) **A separate Patient Information and Informed Consent sheet for blood / tissue samples taken for future testing. The Participant Information Sheet may be incorporated into the consent form, or the consent form may be submitted separately:**
(10) **Please ensure to INVITE the participant to take part in the study; please include a greeting and introduce yourself.**

The HREC requests that the participant be invited – (using a friendly tone):

5 The **Informed Consent Form** should include a clear statement that the participant is consenting to involvement in research, and not to treatment, which will not necessarily provide personal benefit. Any personal benefit should be mentioned when this is possible. In a trial **containing a placebo**, the participant must be made aware that, although the potential risks and benefits of all the substances under trial have been explained, none of the active substances may be administered and it will not be possible for the researcher to reveal whether an active substance or placebo is being administered. An important piece of information is that the participant is free to withdraw from the trial at any time without prejudicing any treatment that is required for existing or future medical conditions. If this is not made clear, the researcher risks the accusation that consent was obtained by subtle coercion (that is, the possibility of prejudice against the participant as a current or future patient).

6 STORAGE OF BLOOD AND/OR TISSUES SAMPLES:
The ethical issues surrounding the storage of blood and/or tissue samples internationally and in South Africa
"If, blood specimens are to be stored for future analysis and it is planned that such analysis will be done outside Wits then the blood must be stored at Wits with release of sub-samples only once projects have been approved by the local Research Ethics Committee applicable to where the research will be done as well as by the Wits Human Research Ethics Committee: (Medical)."

7. SOUTH AFRICAN NATIONAL CLINICAL TRIALS REGISTRY (SANCTR)
The South African Cochrane Centre (SACC), in partnership with the National Department of Health (DoH) and the Cochrane Infectious Disease Group, has established a South African registry for all clinical trials conducted in South Africa

The South African National Clinical Trials Registry (SANCTR) is a South African initiative to support clinical trial registration within the Republic of South Africa. The SANCTR is an official registry and member of the World Health Organization (WHO) Network of Primary Registers. The SANCTR provides data to the central WHO International Clinical Trials Registry Platform (ICTRP) through the regional register - PACTR, so that trials registered in the SANCTR are represented in global searches (WHO ICTRP search portal URL: http://apps.who.int/trialsearch)

Please note that there is a new **South African National Clinical Trials Registry** (SANCTR) database, and that the South African Medical Research Council and the NHREC system/platform is no longer in use. Going forward the only point of trial data entry is the new SANCTR database on this url: https://sanctr.samrc.ac.za/. Please follow this link for information on how to register.